

CHOICE OF EXPERTS IN TRANSFORMER MODELS

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ABSTRACT

As generative AI models grow ever larger in size, the cost of training and inference has grown with them. Sparely activated models, however, have the potential to massively increase the number of trainable parameters while keeping training and inference costs low. A common technique used in such models is Mixture of Experts (MoE), where a model selects different parameters depending on the input. Most applications of MoE to the transformer model commonly used in language tasks use the feedforward part of the network as the expert. However, it is also possible to use other parts of the transformer model as experts. We apply this technique on various parts of the BERT language model, including feedforward, self-attention, and the entire transformer, and compare their efficacy on the Multi-Genre Natural Language Inference (MNLI). We find that all models train and inference at similar rates despite the MoE models having much more trainable parameters, but that nevertheless the MoE models do not achieve improved expressiveness on our dataset.

1 INTRODUCTION

There has been much activity in the generative AI sector recently, as models like ChatGPT and Stable Diffusion have received mainstream use and recognition. However, as these models have grown in capability, they have also grown ever larger in size, and the cost of training and inference has grown with them. GPT3, OpenAI’s last-gen model, had 175 billion parameters and took $3.14\text{E}+23$ flops of compute to train (Brown et al., 2020). Their newest model, GPT4, had 1.76 trillion parameters (OpenAI, 2023), a tenfold increase.

These prohibitive costs have led to recent interest in sparsely activated models which have the potential to massively increase the quantity of trainable parameters while maintaining training and inference costs. One such model is Mixture of Experts (MoE). This concept revolves around partitioning a network, usually the feedforward network, into numerous “expert” sub-networks, each specializing in different parts of the problem at hand. Then through a gating network, which is also a trainable component, specific inputs are assigned to the most suitable expert. This decision process is typically managed by a softmax function, which assigns probabilistic weights to each expert for a specific input, determining their contribution level to the final output. The gating network, besides assigning inputs to experts, also fine-tunes their training by learning the optimal expert for each input type during the training phase, using gradient-based optimization methods similar to standard neural network training (Shazeer et al., 2017).

MoE models differ from traditional neural networks in parameter activation. Unlike conventional models that use all parameters for every input, MoE models activate only a subset of parameters based on the input, allowing for parameter size scaling without a proportional increase in computational costs (Shazeer et al., 2017).

1.1 PROBLEM

In the context of transformers, MoE layers have commonly used the feed-forward block (FFN) within the transformer as the expert, as in the Google Switch Transformer (Fedus et al., 2022). However, this technique could also reasonably be applied to the self-attention block (ATTN), or to

the transformer as a whole. We apply this technique on these parts of the Bidirectional Encoder Representations from Transformers (BERT) model (Devlin et al., 2019) and compare their efficacy.

1.2 EVALUATION

We compare these experts in through training cost, inference cost, number of parameters, and accuracy. Our benchmark for this comparison is the Multi-Genre Natural Language Inference (MNLI) corpus Williams et al. (2018), a dataset of two-sentence pairs which are labeled as “entailment,” “neutral,” or “contradiction.” Table 1 provides some examples of such pairs.

Sentence 1	Label	Sentence 2
At 8:34, the Boston Center controller received a third transmission from American 11.	entailment	The Boston Center controller got a third transmission from American 11.
8 million in relief in the form of emergency housing	neutral	The 8 million dollars for emergency housing was still not enough to solve the problem.
Met my first girlfriend that way.	contradiction	I didn’t meet my first girlfriend until later.

Table 1: Caption

We finetune and test on this dataset and take time for finetuning as training cost, time for testing as inference cost, and accuracy on the test data as accuracy.

2 RELATED WORK

Our primary reference in this work was Outrageously Large Neural Networks: The Sparsely Gated Mixture-of-Experts Layer by Shazeer et al. (2017). Our MoE implementation follows theirs, which works as follows.

2.1 MIXTURE OF EXPERTS

A MoE layer contains n “expert networks” $E_1, \dots, E_n$ and a gating network G whose output is a sparse n -dimensional vector. Then the output of the MoE module is

$$y = \sum_{i=1}^n G(x)_i E_i(x).$$

Because G is sparse, computation is saved as $E_i(x)$ need not be computed when $G(x)_i = 0$. We utilize Noisy Top-K Gating

$$\begin{aligned} G(x) &= \text{Softmax}(\text{KeepTopK}(H(x), k)) \\ H(x)_i &= (x \cdot W_g)_i + \text{StandardNormal}() \cdot \text{Softplus}((x \cdot W_{\text{noise}})_i) \\ \text{KeepTopK}(v, k)_i &= \begin{cases} v_i & \text{if } v_i \text{ is in the top } k \text{ elements of } v \\ -\infty & \text{otherwise.} \end{cases} \end{aligned}$$

As discussed in Shazeer et al. (2017), the noise term helps with load balancing, where W_{noise} is a trainable parameter that controls for the amount of noise.

2.2 RELATED MODELS

Other related work includes MoEBERT (Zuo et al., 2022), where the FFNs are used as the experts and compared against other optimized versions of BERT models. Here an importance score, originally introduced for model pruning, is used to reduce the number of parameters in each FFN expert. However, because of the more complex structure of the ATTN and transformer blocks which we would like to use as experts, we were not able to replicate this technique for our application.

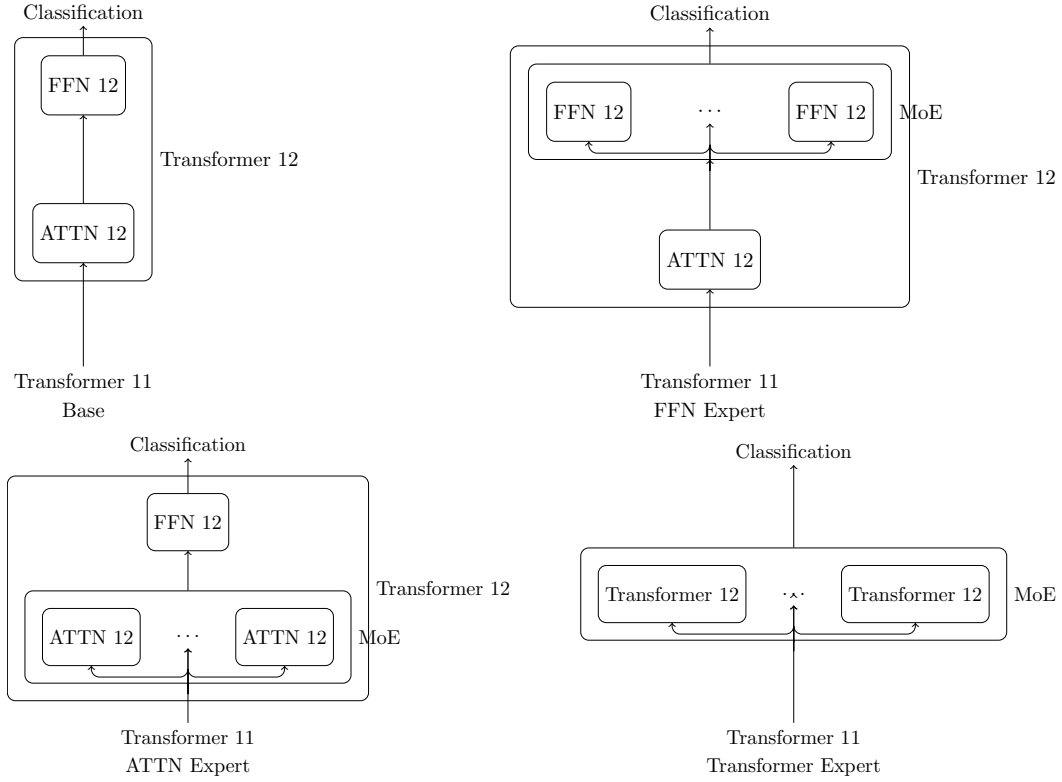


Figure 1: Diagrams of the Models

3 METHODS

We take BERT as our base model and utilize the MoE structure given in Outrageously Large Neural Networks (Shazeer et al., 2017), but we make some modifications to this model. Firstly, the BERT Embedding layer converts the input into a three dimensional tensor of size $b \times n \times 768$, while the gate as implemented in Shazeer et al. (2017) is only capable of taking in a $b \times n$. As such, we flatten before dispatching and unflatten when passing through the expert. Also, the MoE as implemented only dispatches an input tensor, whereas for ATTN and transformer we need to dispatch attention masks as well, which we do using the same assignment given to the inputs. We take $k = 1$ for our Top-K Gating in order to minimize compute cost.

3.1 COMPARISON

We compare the effectiveness of using various elements of BERT as experts, including the FFN, ATTN, and transformer block. Because more general features are captured in the earlier stages of the model, we specifically alter the last transformer in the model. We create copies of these blocks and insert them as experts in the MoE Layer, and then replace the original block with the MoE Layer. The diagrams in Figure 1 depict the type of comparison we wish to make.

In order to engage in such a comparison, we hold the number of parameters constant. To do so, we use 8 copies of the FFN block, 10 copies of the ATTN blocks, and 4 copies of the entire transformer. In a classification task with 3 categories, the base model has 109M trainable parameters and we result in 132M trainable parameters with the FFN as the expert, 133M trainable parameters with the ATTN as the expert, and 131M with the transformer as the expert.

4 EXPERIMENTS

We examine the models’ performance and training and inference time when finetuning on the MNLI dataset. Because MNLI is used as a competitive benchmark, labeled test data is not publicly available, and performance is instead computed when uploading submissions to the competition page on Kaggle. Because only 2 submissions are permitted per day, for the sake of time, we simply take a random 80-20 split of the training data as our train and test datasets. We utilize the AdamW (Loshchilov & Hutter, 2019) optimizer with an initial learning rate of $2e - 5$ and a linearly decreasing scheduler. We train on a laptop RTX 4070 which limits us to a batch size of 48 with each observation truncated to 128 tokens. We notice that loss tends to stabilize around halfway through the first epoch, so we train for only one epoch.

Expert	Parameters	Training Time	Test Time	Accuracy
Base	109,484,547	48m 42.2s	4m 5.7s	81.79%
FFN	132,296,451	50m 36.8s	4m 7.0s	81.85%
ATTN	132,725,763	50m 24.3s	4m 6.2s	81.75%
Transformer	131,534,595	49m 53.6s	4m 8.6s	81.72%

Table 2: MNLI Results

4.1 RESULTS

Our basic results are summarized in Table 2. As can be seen, although the mixture of experts approach adds a significant number of trainable parameters, increasing model size by almost 30%, training and inference time are only slightly higher, and not negatively effected to a significant degree. However, despite the larger parameter count, we also did not achieve much higher accuracy on the modified models when compared to the base model.

	Base			FFN			support
	precision	recall	f1-score	precision	recall	f1-score	
neutral	0.79	0.78	0.78	0.80	0.77	0.78	26110
entailment	0.83	0.85	0.84	0.83	0.85	0.84	26223
contradiction	0.83	0.83	0.83	0.83	0.84	0.83	26208
accuracy			0.82			0.82	78541
macro avg	0.82	0.82	0.82	0.82	0.82	0.82	78541
weighted avg	0.82	0.82	0.82	0.82	0.82	0.82	78541

	ATTN			Transformer			support
	precision	recall	f1-score	precision	recall	f1-score	
neutral	0.79	0.77	0.78	0.79	0.77	0.78	26110
entailment	0.83	0.85	0.84	0.83	0.85	0.84	26223
contradiction	0.83	0.83	0.83	0.83	0.84	0.83	26208
accuracy			0.82			0.82	78541
macro avg	0.82	0.82	0.82	0.82	0.82	0.82	78541
weighted avg	0.82	0.82	0.82	0.82	0.82	0.82	78541

Table 3: `classification_report` from `sklearn.metrics` for each result

More detailed results can be seen in Table 3. As can be seen, the various modifications of the model all seem to achieve a similar precision, recall, and F_1 -score on the dataset. As such, while we have successfully increased the number of trainable parameters in the model with the mixture of experts technique, we do not appear to have increased the model’s expressiveness. Indeed, throughout training, all models appear to have converged to a per-batch loss of around 0.5 within the first 1000 batches, with little further improvement through the rest of the dataset.

5 CONCLUSION

We find that the MoE models succeed in achieving a much higher number trainable parameters while maintaining similar training and inference speeds. Nevertheless, despite their increased parameter count, the MoE models do not achieve improved expressiveness on our dataset.

5.1 LIMITATIONS

There are several reasons for which we believe this may be the case, all of which are related to the expressiveness of the experts. Firstly, due to the single epoch of training, which finetuned the whole model, there may not have been enough time for the experts to sufficiently diverge. A second epoch, only training the experts with the rest of the model frozen, may have increased the overall expressiveness of the experts, and would also have run much faster. Secondly, taking only the top expert forced each expert to be able to completely process each input, and may have limited the ability of the experts to specialize. Thirdly, we may simply not have had enough experts, with only 4 transformers, 8 FFNs, and 10 ATTNs, which may also have limited specialization. Unfortunately, due to our lack of computing resources, we were unable to resolve these issues within the allotted timeframe, but they act as potential avenues for further exploration.

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